

FORMATTING INSTRUCTIONS FOR ICLR 2027

CONFERENCE SUBMISSIONS

Anonymous authors

Paper under double-blind review

ABSTRACT

1 INTRODUCTION

2 RELATED WORKS

3 PRELIMINARIES

3.1 MULTI-AGENT COORDINATION AS BELIEF-CONDITIONED DECISION MAKING

Multi-agent coordination requires an agent to anticipate how other agents will act when selecting its own action, especially when their objectives, incentives, or relevant information are not directly observable. Prior work on opponent modeling addresses this challenge by representing other agents with explicit behavioral models and inferring their actions from interaction histories (Carmel & Markovitch, 1996). Existing work formalize this setting as an interactive partially observable Markov decision process (I-POMDP), in which the interactive state includes models of other agents and observations update beliefs over those models (Gmytrasiewicz & Doshi, 2005).

Interactive model. Following the I-POMDP definition, the direct N -agent form for agent i is

$$\text{I-POMDP}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle, \quad IS_i = S \times \prod_{j \neq i} M_j, \quad A = \prod_{j \in N} A_j, \quad (1)$$

[TODO: put the formal definition in the Appendix] where S is the physical state space, M_j is the model space for agent j , A is the joint action space, and T_i , O_i , and R_i are the transition, observation, and reward specifications. In the intentional model case, a model of agent j is a type $\theta_j = \langle \tilde{b}^j, \hat{\theta}_j \rangle$, where $\tilde{b}^j$ is the belief state attributed to j and $\hat{\theta}_j$ is its frame, including its observation, action, transition, reward, and optimality specifications. Throughout the remainder of this section, $b_t^i \in \Delta(IS_i)$ denotes only the time-indexed belief maintained by agent i over its interactive state space; the tilde-marked $\tilde{b}^j$ is used only for a belief nested inside a hypothesized model of another agent. Under model non-manipulability (MNM), agent i 's actions do not directly modify agent j 's model. Under model non-observability (MNO), agent i cannot directly inspect that model. Accordingly, beliefs about other-agent models can change only through observations generated by the environment and by other agents' behavior, although an agent may still influence those observations indirectly through the public state.

The belief-state estimator is the Bayesian update operator τ_i . Given agent i 's current belief, action, and subsequent observation, it propagates the belief through the physical transition and the action models of the other agents, conditions on the observation likelihood, and normalizes the result:

$$b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i), \quad (2)$$

For intentional models, the action prediction may itself invoke the belief update specified by the modeled agent. For a candidate action a_t^i , let $\pi_{-i}(\cdot \mid x, a_t^i)$ denote the joint response distribution induced by the other-agent models at interactive state x . The belief-averaged immediate reward is the expectation

$$\bar{R}_i(b_t^i, a_t^i) = \mathbb{E}_{\substack{x \sim b_t^i \\ a_{-i} \sim \pi_{-i}(\cdot \mid x, a_t^i)}} [R_i(x, (a_t^i, a_{-i}))]. \quad (3)$$

Here, “averaged” means expectation under the current belief and the response distribution predicted by the supported models, rather than an arithmetic average. This is the standard belief-state notation for the immediate reward; the same predictive distribution induces the observation probabilities in the standard belief-state Bellman recursion below:

$$V_i(b_t^i) = \max_{a_t^i \in A_i} \left[\bar{R}_i(b_t^i, a_t^i) + \gamma \sum_{o_{t+1}^i} \Pr(o_{t+1}^i \mid b_t^i, a_t^i) V_i(\tau_i(b_t^i, a_t^i, o_{t+1}^i)) \right]. \quad (4)$$

Partner belief and conditioned planning. [TODO: I still think partner is not an ideal choice here, we need a more formal one] We call the portion of b_t^i that concerns the behaviorally relevant components of M_j a *partner belief*. Such a component may encode a hidden goal, incentive, behavior, or relevant knowledge of agent j . *Partner-conditioned planning* is the decision making operation that uses this belief. For a candidate action a_t^i , agent i evaluates the responses of the other agents under the models supported by b_t^i , applies the resulting state transition, and accounts for downstream outcome before selecting an action. The action may be a task action, a proposal addressed to a particular agent, or a response to an offer. In this sense, the belief determines the partner model used in planning, while planning determines which action is taken and hence which responses and observations can occur. [TODO: I have edited this part and you should read my framing and wording]

The two operations therefore form a closed sequential dependency:

$$b_t^i \xrightarrow[\text{response models and utility}]{\text{prediction and evaluation}} a_t^i \xrightarrow[\text{environment and other-agent transition}]{\text{prediction and evaluation}} o_{t+1}^i \xrightarrow{\tau_i} b_{t+1}^i. \quad (5)$$

[TODO: I have no idea what has been written around the arrow, this is awkward, replace them] Equation equation 5 is a functional decomposition of the standard I-POMDP mapping.

3.2 BENAC-P ENVIRONMENT AND ORACLE PARTNERS

[TODO: say something like ”to explore on the two ability mentioned above, we build a new game based on BENAC”] It retains the BENAC commitment-based structure: players negotiate irreversible commitments that jointly determine which public goals are achieved, and utility is evaluated from the terminal goal configuration (Benac et al., 2026). The implementation used in this work makes the information structure explicit by assigning each player a private preference row over the public goals. It also supports an optional two-offer menu.

Game specification. A game instance is summarized by [TODO: again why the K here looks different in the tuple??]

$$\mathcal{M} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \rho, C_0, \mathcal{L}, \mu_0 \rangle. \quad (6)$$

Here $\mathcal{N} = \{0, \dots, N-1\}$ is the player set, and $\mathbf{K} = (K_i)_{i \in \mathcal{N}}$ records the number of commitment coordinates owned by each player. The public goals are indexed by $\mathcal{G} = \{0, \dots, G-1\}$. For each g , the requirement set Γ_g contains action references (i, k) with $0 \leq k < K_i$. The schedule $\rho = (\rho_0, \dots, \rho_{T-1})$ specifies the proposer at each of the T proposal turns, C_0 is the all-zero initial commitment state, and $\mathcal{L}$ denotes the state-dependent legal-action relation. The symbol μ_0 denotes the declared prior over private preference configurations.

[TODO: I have deleted all explnaction from you of the code, do not explain any thing related to code, this is awkward]

Commitments, goals, and utility. At turn t , the commitment state is $C_t = (c_{t,i})_{i \in \mathcal{N}}$ with $c_{t,i} \in \{0, 1\}^{K_i}$. Commitments are irreversible, which means one entry cannot be unset by a later action. With $\delta_g \in \{\text{bin}, \text{lin}\}$ denoting the goal mode, its satisfaction is

$$S_g(C_t) = \begin{cases} \mathbf{1}[c_{t,i,k} = 1 \text{ for every } (i, k) \in \Gamma_g], & \delta_g = \text{bin}, \\ \frac{1}{|\Gamma_g|} \sum_{(i,k) \in \Gamma_g} c_{t,i,k}, & \delta_g = \text{lin}. \end{cases} \quad (7)$$

Each goal involves at least two players. The environment samples a private preference matrix $V = (v_{i,g})$ with $v_{i,g} \in \{-1, 0, +1\}$, corresponding to AVOID, NEUTRAL, and WANT. Player i observes only its own row v_i . At termination, after all turns in ρ have been consumed, its utility is $U_i(C_T; v_i) = \sum_{g \in \mathcal{G}} v_{i,g} S_g(C_T)$ [TODO: explain use one sentence how the utility is given]

Native actions and transitions. Let $p = \rho_t$ be the current proposer and let $s_t = (C_t, t, H_t)$ denote the public physical state, turn index, and public transcript. A proposer may choose PASS or an ordinary offer to any $q \neq p$. An offer is $o = (q, \bar{c}_p, \bar{c}_q)$. Existing commitments must remain set; each side may add at most M new commitments; forbidden commitments may not be added; and an offer must add at least one commitment in total.

For an ordinary offer, the addressed player chooses ACCEPT or REJECT. Acceptance binds the proposed commitments immediately:

$$c_{t+1,i} = \begin{cases} c_{t,i} \vee \bar{c}_i, & i \in \{p, q\}, \\ c_{t,i}, & \text{otherwise,} \end{cases} \quad (8)$$

whereas rejection leaves C_t unchanged. A completed proposal–response interaction is recorded in the public transcript and advances the schedule by one proposer turn. PASS advances the schedule immediately without changing commitments.

[TODO: delete the separate menu paragraph and integrate it into the explanation above] When `menu_enabled` is true, the proposer may instead submit a menu $m = (o^{(1)}, o^{(2)})$ containing two distinct legal offers addressed to the same player. The responder then chooses CHOOSE_1, CHOOSE_2, or REJECT; exactly the selected offer binds, and rejection binds neither. Menus are the only additional proposal form: enabling them adds a MENU proposal and CHOOSE_1/CHOOSE_2 responses to the core PASS/OFFER and ACCEPT/REJECT actions. Offers, responses, resulting commitments, and the transcript are public. A pending offer is not binding until the responder accepts it or selects one of its options.

[TODO: the blow 3 paragraph is really hard to understand. you should not mention anything about diagnose experiment here yet, and what you need to do is to show that how we build oracle player, how we update belief and planning for oracle player, show that use formula, and add MCTS...]

Belief filtering. For agent i , let $h = (v_j)_{j \neq i}$ be a joint hypothesis about the other players' private rows. Let $\mathcal{V}_j$ denote the public finite catalogue of candidate rows for player j ; in the native diagnostic, $\mathcal{H}_t^i \subseteq \prod_{j \neq i} \mathcal{V}_j$ is the currently retained support. A declared prior is conditioned on v_i and any supplied initial support. If agent i observes a public action u_t^j by player j (a proposal or a response), the general finite-support update is

$$b_{t+1}^i(h) = \frac{b_t^i(h) \ell_j(u_t^j | I_t^j, h)}{\sum_{h' \in \mathcal{H}_t^i} b_t^i(h') \ell_j(u_t^j | I_t^j, h')}, \quad h \in \mathcal{H}_t^i, \quad (9)$$

where I_t^j contains the information available to the acting player and ℓ_j is the likelihood induced by its declared policy. A deterministic policy gives $\ell_j(u | I_t^j, h) = \mathbf{1}[f_j(I_t^j, h) = u]$, so the update reduces to support filtering,

$$\mathcal{H}_{t+1}^i = \{h \in \mathcal{H}_t^i : f_j(I_t^j, h) = u_t^j\}.$$

For a kernel that depends only on the acting player's row, the likelihood reduces to a function of h_j . The active oracle used in the native diagnostic may also depend on its inferred distribution over the other rows; in that case the dependence on the full h in equation 9 captures the acting player's own information state. The solver treats agent i 's own actions as interventions rather than as likelihood evidence. A public commitment transition is used to check the history and update the physical state; it does not by itself reveal a private row. The repository's general `ExactBayesFilter` implements the weighted version for known `SoftProgressPolicy` kernels, while the native diagnostic uses the deterministic support form over explicit, independent per-player catalogues. Thus, in the latter setting, any cross-player dependence in a posterior comes from observed public behavior or conditioning, not from a shared type label.

Oracle partners. The native diagnostic uses `RationalPartner` as an information-limited oracle. At a proposal or response decision, player j is supplied with its public information $I_t^j = (s_t, H_t, \text{pending offer}, v_j, \mathcal{A}_t^j)$ and chooses from the complete legal native action set according

to

$$\pi_j^{\text{orc}}(u \mid I_t^j) \in \arg \max_{u \in \mathcal{A}_t^j} \mathbb{E} \left[U_j(C_T; v_j) \mid I_t^j, u, \pi_{-j}^{\text{ref}} \right]. \quad (10)$$

The expectation is over the other players’ independent public type catalogues, conditioned on j ’s own row and the public history. The oracle enumerates the remaining original schedule, propagates the exact binding commitment transitions, and evaluates terminal utility. Its fixed reference continuation uses the code’s immediate rule: a responder chooses the action with the greatest currently satisfied own-goal utility, with rejection winning ties; a proposer chooses the ordinary offer with greatest immediate own utility under acceptance, with PASS winning ties. The reference rule does not propose menus, although the terminal best-response search enumerates all legal offers, menus, and responses.

Past actions by oracle players are replayed under their actual oracle policy when a counterfactual information state is constructed; actions taken by the evaluated player are interventions and are not treated as evidence about its own hidden row. If a policy requires an inference about another player, that inference is recomputed from the acting player’s public history and catalogue, not replaced by the environment’s realized private information. The implementation has no stochastic action error. The resulting policy is an exact best response to the declared reference continuation, not an equilibrium or a full-information solution. The legacy `PerfectInfoSolver` is a separate full-information mechanism-checking tool and is not the oracle used for menu-enabled native endgames.

Reference planning. At a decision of player i , the exact search enumerates the legal action set for the current role, simulates the intervening oracle actions, updates the compatible support, and replans at the next decision of player i . Let e denote the intervening public evidence, s_e the resulting physical state, and τ_i the update in equation 9. The finite-horizon reference recursion is

$$W_i(s, H, b) = \max_{a \in \mathcal{A}_i(s, H)} \sum_e p_b(e \mid s, H, a) W_i(s_e, H \oplus (a, e), \tau_i(b, H, a, e)), \quad (11)$$

$$W_i(s_T, H, b) = U_i(C_T; v_i).$$

Here $\mathcal{A}_i(s, H)$ is a proposal set when i proposes and a response set when i is responding. Response decisions are therefore not folded into an automatic approval step. The native endgame search stops a branch at the next such decision or at termination, while all intervening partner actions retain their probabilities or deterministic support weights. The planner optimizes terminal utility under the specified partner policy; it does not optimize information gain as a separate objective and does not access the realized private preference rows.

4 DIAGNOSE

We diagnose two capabilities within the interaction loop $B_t \rightarrow P_t \rightarrow e_{t+1} \rightarrow B_{t+1}$: forming a task-relevant partner judgment (B) and using it to plan an interaction (P). The diagnosis separates two questions: *what dependencies does the task impose?* and *can the model perform the computations required at each interface?* An unreliable planner can fail to use a correct judgment, and an unreliable updater can fail to use informative evidence. Consequently, the final model output alone cannot distinguish an absent dependency from a failure to exploit it. We first certify both directions with an oracle, then measure model errors under controlled judgments and evidence. The effect of replacing one model component is a supplementary observation, not the criterion for whether a task dependency exists.

4.1 NATIVE ENDGAMES AND PAIRED INTERVENTIONS

Controlled positions. We freeze legal public histories in three-player BENAC-P games and measure at most two actual ego decisions, including any intervening response decision. The original proposal schedule, full legal action set, independent preference catalogues, and irreversible commitments are retained. Partners actively propose and respond using the history-aware deterministic policy in Section 3.2. In each position, one partner-goal preference is queried; all other preference entries are supplied, and uncertainty is over WANT, NEUTRAL, and AVOID. Exact search supplies reference judgments and terminal-utility action values, without exposing the realized hidden preference to ego.

We use two oracle-selected cohorts of six positions each, sharing two positions: ten positions from ten source games in total. The $B \rightarrow P$ cohort requires an action that is optimal under one supported partner type but strictly suboptimal under another. Common optimal actions are allowed and remain correct. The $P \rightarrow B$ cohort requires two legal first actions that induce different expected posterior uncertainty and leave a next ego decision on every response branch. These forced actions need not be utility-optimal; the reference action used for planning regret always maximizes terminal utility over all legal actions. Selection uses game mechanics and oracle values, not model errors or the sign of a repair effect.

A concrete negotiation. Consider `native-30040-g6-d2`, where ego is P_1 , all three players have already committed to A_0 , and the remaining proposers are P_1 then P_0 . The queried preference is P_2 's attitude toward $G_6 = (P_0.A_1 \wedge P_2.A_1)$. Ego can offer O_1 alone or the menu $\{O_1, O_2\}$, with new commitments

$$O_1 : P_1 \text{ adds nothing, } P_2 \text{ adds } A_1; \quad O_2 : P_1 \text{ adds } A_1, P_2 \text{ adds } A_1.$$

Under the declared oracle policy and this game's other preferences, all three P_2 types accept O_1 alone, leaving the queried preference unresolved. With the menu, however, AVOID chooses O_1 , whereas WANT and NEUTRAL choose O_2 . Thus the choice supports either $\{\text{AVOID}\}$ or $\{\text{WANT}, \text{NEUTRAL}\}$, while immediately binding the selected commitments. Next, P_0 actively proposes adding its own A_1 without further commitments from P_1 , who must accept or reject. This is evidence acquisition followed by a native decision, not an explicit query action. In the observed O_1 menu branch, the model reports $\{\text{WANT}, \text{NEUTRAL}\}$ despite the response supporting $\{\text{AVOID}\}$. Appendix A.1 gives the preferences and oracle payoff comparison underlying this example.

Separate capability measurements. For B , the model receives the initial possibilities and public history and returns the set of partner preferences still compatible with that history. We score exact set agreement, allowing a correct answer to remain uncertain. For P , we supply the evidence-supported reference judgment and ask the model to select a legal action. Regret is the difference from the optimal expected terminal utility, with reference continuation after the chosen action; all optimal ties receive zero regret. The model outputs brief reasoning followed by a semantic judgment or action index, never probabilities, Q values, or a plan variable. Since P also receives history, this is planning assisted by a correct judgment, rather than isolation of an internal planning module.

Certifying dependencies and probing their use. For $B \rightarrow P$, exact search holds the physical state, public history and legal actions fixed and compares action values under singleton judgments for two still-compatible partner types. An action optimal under one judgment must be strictly sub-optimal under the other. This certifies that partner understanding affects action evaluation; shared optimal actions remain valid. For $P \rightarrow B$, we force two legal first actions from the same root, enumerate subsequent partner actions up to the next ego decision, and compute the evidence-supported judgments and their expected uncertainty. This certifies that interaction choice affects what can subsequently be inferred. Both are oracle selection criteria, established independently of model performance.

We then ask the model to form judgments from those histories and to plan with the evidence-supported reference judgment. These tests locate errors in B and in reference-judgment-assisted P , respectively. As a supplementary $B \rightarrow P$ probe, we replace the model's explicit judgment with the reference judgment in otherwise identical fresh planning contexts. The replacement supplies only what the evidence supports, not the realized hidden type. Action changes and signed regret differences measure the model's response to this input intervention; neither a positive repair benefit nor a gain in accuracy between evidence arms is required to certify the task dependencies. Because actions also change commitments, utility contrasts are not interpreted as pure information effects.

4.2 CERTIFIED DEPENDENCIES AND MODEL ERRORS

We evaluate Qwen3-4B-Instruct-2507 at temperature zero. All 127 diagnostic tasks returned valid submissions, with no truncation and a mean of 230.8 completion tokens. The separate empty-history controls scored 3/4: a known WANT preference was incorrectly expanded to all three possibilities. We retain this error and therefore do not attribute all subsequent belief errors exclusively to strategic inference. The cohorts are development diagnostics, not an untouched confirmation set. Table 1

Table 1: Two levels of diagnosis. Panel A reports exact task properties required for selection, not model success rates. Panel B measures model performance at the corresponding interfaces. Each direction uses six source games, with two shared across directions. Intervals are descriptive 95% game-bootstrap intervals; regret uses additive goal-utility units.

Diagnostic quantity	Result	95% interval
<i>A. Oracle-certified task dependencies</i>		
$B \rightarrow P$: judgment changes action optimality	6/6 cases	—
$P \rightarrow B$: action changes inferable preferences	6/6 cases	—
Posterior entropy: lower / higher information (bits)	1.585 / 0.667	—
<i>B. Model errors at the two interfaces</i>		
Root B exact, judgment-sensitive cohort	16.7%	[0.0, 50.0]%
P regret with oracle judgment, judgment-sensitive cohort	0.444	[0.167, 0.722]
P regret with oracle judgment, action-evidence cohort	1.000	[0.556, 1.556]
B exact after forced lower-information action	16.7%	[0.0, 50.0]%
B exact after forced higher-information action	22.2%	[5.6, 44.4]%

separates oracle-certified dependencies from model measurements; each direction uses its own six-game cohort. Appendix A gives selection, execution and aggregation details.

The task couples partner understanding and interaction choice. Panel A makes the two dependencies explicit. In every judgment-sensitive case, changing the assumed partner type changes the optimality of at least one legal action. In every action-evidence case, changing the forced interaction changes the set of partner preferences that subsequent behavior supports. The latter reduces expected oracle uncertainty by 0.918 bits for each selected pair. Thus, both dependencies exist even before considering whether the model can compute an appropriate judgment or action. These are properties of the selected endgames, not estimates of their prevalence in random games.

The model makes errors at both interfaces. Panel B tests the model against those oracle-defined requirements. At the judgment-sensitive roots, only one of six reported partner judgments is exact. When the reference judgment is supplied, planning regret remains positive: 0.444 in the judgment-sensitive cohort and 1.000 in the action-evidence cohort. The planning test therefore does not require the model first to infer the correct judgment unaided. Conversely, the forced-action tests provide the corresponding evidence independently of the model’s first-action choice. Judgment accuracy is 16.7% under the lower-information action and 22.2% under the higher-information action. A correct unresolved set receives full credit; these errors concern agreement with what the evidence actually supports, not failure to identify a type that remains ambiguous.

Joint responses are supplementary, not a dependency test. Replacing model judgments changes three of six root actions in the judgment-sensitive cohort, with two utility-neutral changes and one decrease in utility. The mean regret reduction is -0.111 ; the paired difference in B accuracy between the forced evidence arms is 5.6 percentage points, with an interval spanning zero. Appendix A reports these signed effects. They describe how the evaluated model responds to the interventions. An output that remains wrong after receiving a correct judgment or more informative evidence does not erase the independently certified task dependency.

The diagnosis therefore identifies errors in both partner judgment and partner-conditioned planning within tasks that couple the two computations. It separates the existence of those dependencies from the model’s ability to use them. This motivates training B and P as separate targets and then evaluating their composition and transfer. It does not identify internal neural modules or establish that the two errors causally amplify one another.

5 TRAINING

6 TEST AND TRANSFER

7 DISCUSSION

8 CONCLUSION

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, “we checked LLM-generated research ideas for potential plagiarism through a manual literature survey”, “LLM-generated code was verified and tested for correctness by 2 authors”, etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

See the ICLR 2027 AI Policy for Authors for more details. This statement should not be more than 1 page.

ETHICS STATEMENT

(This section is **recommended** and does not count toward the page limit.)

If authors feel that their paper submission raises questions regarding the Code of Ethics, they are encouraged to include a paragraph of Ethics Statement (at the end of the main text before references) to address potential concerns where appropriate. Topics include, but are not limited to, studies that involve human subjects, practices to data set releases, potentially harmful insights, methodologies and applications, potential conflicts of interest and sponsorship, discrimination/bias/fairness concerns, privacy and security issues, legal compliance, and research integrity issues (e.g., IRB, documentation, research ethics). This statement should not be more than 1 page.

REPRODUCIBILITY STATEMENT

(This section is **recommended** and does not count toward the page limit.)

It is important that the work published in ICLR is reproducible. Authors are strongly encouraged to include a paragraph-long Reproducibility Statement at the end of the main text (before references) to discuss the efforts that have been made to ensure reproducibility. This paragraph should not itself describe details needed for reproducing the results, but rather reference the parts of the main paper, appendix, and supplemental materials that will help with reproducibility. For example, for novel models or algorithms, a link to an anonymous downloadable source code can be submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement.

AUTHOR CONTRIBUTIONS

If you’d like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

REFERENCES

- Leo Benac, Jonas Raedler, Zilin Ma, and Finale Doshi-Velez. A benchmark for multi-party negotiation games from real negotiation data. *arXiv preprint arXiv:2603.14066*, 2026. URL <https://arxiv.org/abs/2603.14066>.
- David Carmel and Shaul Markovitch. Opponent modeling in multi-agent systems. In *Adaptation and Learning in Multi-Agent Systems*, volume 1042 of *Lecture Notes in Computer Science*, pp. 40–52. Springer, 1996. doi: 10.1007/3-540-60923-7_18.
- Piotr J. Gmytrasiewicz and Prashant Doshi. A framework for sequential planning in multi-agent settings. *Journal of Artificial Intelligence Research*, 24:49–79, 2005. doi: 10.1613/JAIR.1579. URL <https://www.cs.cmu.edu/afs/cs/project/jair/pub/volume24/gmytrasiewicz05a.pdf>.

A NATIVE DIAGNOSTIC PROTOCOL AND SUPPLEMENTARY RESULTS

Positions and oracle selection. Each frozen position contains its original game specification, independent per-player preference catalogues, ego’s preference row, a queried partner–goal pair and a replayable legal history. The ten positions have three players, two binary commitments per player, eight public goals and two scheduled proposal turns per player. At most three original proposal turns remain at the root. No preparation, assessment or automatic approval action is introduced. The active partner policy uses public history and its own preferences; its terminal best response is to the declared fixed continuation, not to an omniscient or equilibrium opponent.

For the judgment-sensitive cohort, the oracle evaluates all legal actions under each still-supported partner type. Admission requires an action optimal under one type with positive regret under another. This witnesses a consequential dependence on partner understanding, while allowing actions that are optimal under multiple types. It neither requires the model’s judgment to be wrong nor guarantees a change when that particular judgment is repaired. For the action–evidence cohort, native enumeration selects the first pair of legal actions with different expected posterior entropy and a nonterminal next ego decision on every branch. “Higher” and “lower” information refer only to this pair, not global information extrema. The utility-optimal reference action is selected independently. A positive downstream value from updating a frozen prior is not an admission condition.

Table 2: Frozen native positions. D and C retain the source-seed discovery and confirmation labels used during development; C is not an untouched test set. Each direction has three games in each partition.

Position identifier	Partition	$B \rightarrow P$	$P \rightarrow B$
native-30003-g6-d2	C	yes	
native-30005-g0-d1	C	yes	
native-30009-g2-d2	C	yes	
native-30013-g6-d3	C		yes
native-30016-g5-d1	D	yes	yes
native-30018-g5-d2	D	yes	yes
native-30019-g2-d1	C		yes
native-30020-g5-d1	D	yes	
native-30041-g6-d1	C		yes
native-30040-g6-d2	D		yes

A.1 WORKED MENU EXAMPLE

In native-30040-g6-d2, let x_i denote the already-bound $P_i.A_0$ and y_i denote $P_i.A_1$. Initially all x_i are true and all y_i are false. Table 3 gives the complete game; only v , P_2 ’s preference for G_6 , varies across the three compatible types. The public prefix is P_0 pass; a P_1 menu choosing between binding $(P_1.A_0, P_2.A_0)$ and $(P_1.A_1, P_2.A_0)$, with P_2 choosing the former; P_2 offering to add $P_0.A_0$, accepted by P_0 ; and P_2 offering to add $P_1.A_1$, rejected by ego. The resulting state is the example’s root.

Table 3: Complete goal and preference specification of the worked example. Entries +1, 0 and −1 mean WANT, NEUTRAL and AVOID. Here $v \in \{+1, 0, -1\}$ is hidden from ego.

Goal	Required commitments	P_0	P_1 (ego)	P_2
G_0	$x_0 \wedge y_1$	0	+1	0
G_1	$x_0 \wedge y_1 \wedge x_2$	−1	+1	+1
G_2	$y_0 \wedge x_1 \wedge y_2$	+1	−1	−1
G_3	$y_0 \wedge y_1 \wedge y_2$	0	+1	−1
G_4	$x_0 \wedge x_1$	+1	+1	0
G_5	$x_1 \wedge y_2$	0	−1	+1
G_6	$y_0 \wedge y_2$	+1	+1	v
G_7	$y_1 \wedge x_2$	−1	−1	+1

The menu offers O_1 , adding only y_2 , and O_2 , adding y_1 and y_2 . Exact terminal values for P_2 , evaluated under its declared fixed reference continuation, are

v	REJECT	CHOOSE_1	CHOOSE_2	oracle response
+1	1	1	2	CHOOSE_2
0	0	1	1	CHOOSE_2
−1	0	1	0	CHOOSE_1

For the neutral type, the immediate-reference tie-break selects O_2 : it immediately satisfies P_2 ’s wanted G_1, G_5, G_7 , whereas O_1 satisfies only G_5 . Thus CHOOSE_1 identifies AVOID, and CHOOSE_2 leaves WANT/NEUTRAL. Offering O_1 alone elicits acceptance from all three types. The comparison illustrates how the menu changes the evidence channel without changing the legal game protocol.

After either menu choice, the actual active P_0 offers to add y_0 while asking ego for no additional commitment. After O_1 , ego’s terminal values for reject and accept are both 0. After O_2 , they are 1 and 2 respectively. The difference follows from the bound commitments: adding y_0 after O_1 activates an avoided G_2 and a wanted G_6 , which cancel; after O_2 it also activates the wanted G_3 . The observed model judgment after the O_1 menu branch is $\{\text{WANT}, \text{NEUTRAL}\}$, excluding the only supported type. This example demonstrates the action–evidence contrast and an inference error; its between-action utility difference also reflects physical commitments. It is one of the six action–evidence cases, not an additional replication.

A.2 MEASUREMENT AND EXECUTION DETAILS

Queries and interventions. Let $\mathcal{B}^*(s, H)$ denote the evidence-supported preference set and $\hat{\mathcal{B}}(s, H)$ the model’s reported set. Both paired planning queries receive the same state s , complete public history H , initial possibilities and legal actions. Only the explicit judgment differs. The belief-generation scratchpad is not passed to either fresh planning context. For a supplied judgment $\mathcal{B}$, write the model action as $\hat{a}(\mathcal{B})$ and let $Q^*(a)$ be its exact expected terminal value under the evidence-supported reference judgment. We report

$$\text{Reg}(a) = \max_{a'} Q^*(a') - Q^*(a), \quad \Delta_B = \text{Reg}(\hat{a}(\hat{\mathcal{B}})) - \text{Reg}(\hat{a}(\mathcal{B}^*)). \quad (12)$$

Positive Δ_B means beneficial repair; either sign can accompany an action change. Exact-set B accuracy requires every supported preference and no unsupported preference. Reference beliefs are uniform over compatible types; all probabilities and action values are computed in the backend.

For forced action a , let E_a be the subsequent sequence of observed partner actions before the next ego decision, and let Z be the uncertain partner type. The channel statistic is $\mathbb{E}[H(Z \mid H, a, E_a)]$ in bits, with every branch weighted by its compatible-type mass. The information gap subtracts this statistic for the higher-information arm from that for the lower-information arm. Model exactness is scored against the appropriate support on each branch, then averaged with the same weights. Terminal branches in other action arms require only a judgment; no fictitious continuation action is added. After the second measured ego action, the reference planner controls the remaining continuation.

Execution and semantic controls. We use Qwen3-4B-Instruct-2507 with temperature zero, served through vLLM 0.28 and the Hermes tool-call parser. The prompt requests brief reasoning with soft targets of 120 words for judgments and 240 for actions. Each response exposes one

submission tool with automatic tool choice and a 1,024-token completion cap. Only truncation permits one additional 128-token submission request, without corrective feedback. All 127 formal tasks were valid on their first request; no finalization was needed. All contained assistant text before the tool call, averaging 131.4 whitespace-delimited words and 230.8 completion tokens per task (29,308 completion tokens in total). These traces are observable explanations, not a validated measure of the model’s internal computation.

Four separate empty-history controls supply singleton WANT, NEUTRAL, AVOID, or the unresolved three-label set. All four returned valid submissions, but the WANT control incorrectly returned all three labels. Its explanation incorrectly described the supplied initial set as containing all three values. The run initially stopped under the semantic gate and then continued under a protocol-validity gate, reusing all four cached responses, including this error. Neither prompts nor inference budgets were changed, and incorrect semantic answers were not retried. The controls consumed 491 completion tokens in four requests, separately from formal-task counts. This failure limits attribution of B errors: some can reflect reading or prior-grounding failures rather than strategic inference alone.

Aggregation and partition results. Panel A of Table 1 counts the six certified positions in each cohort. These counts summarize admission criteria, not six successful model interventions. The judgment-sensitive certificate uses two compatible type-conditioned singleton judgments at the same public state and history; it does not claim that the actual history identifies either singleton. The model’s oracle-judgment input instead contains the full evidence-supported set. Panel B reports model errors under the controlled inputs, separately from these task-level certificates.

The main table combines the two development partitions *within* each six-game cohort. It never pools the overlapping cohorts as twelve independent games. Branches are weighted within games before computing game means. The 95% intervals use 2,000 percentile-bootstrap resamples of source games with random seed zero. They describe variation in this selected small set, not failure prevalence across random games or an untouched confirmatory test. The 0.918-bit channel gap is identical in all six selected pairs; its absence of between-game variation is a property of these examples, not a precision claim about other environments.

Table 4: Results by retained development partition, three games per cell. Regret and judgment repair use the cohort roots. Forced-arm B accuracy is branch-weighted. Judgment replacement changes one action in C and two in D.

Measure	D	C
<i>Judgment-sensitive cohort</i>		
B exact	0.0%	33.3%
P regret with reference judgment	0.389	0.500
Judgment repair: reduction in regret	−0.222	0.000
<i>Action-evidence cohort</i>		
Root P regret with reference judgment	0.667	1.333
B exact, lower-information action	0.0%	33.3%
B exact, higher-information action	11.1%	33.3%

Task consequences and secondary decompositions. The following joint-response measurements supplement the oracle certificates and interface-specific error measurements; their sign is not an admission criterion or a test of whether the task dependency exists. Across the six judgment-sensitive cases, mean model-planner regret reduction is -0.111 $[-0.333, 0.000]$. Across the six action-evidence cases, the paired difference in B exactness is 5.6 $[-38.9, 44.4]$ percentage points. Neither quantity establishes a positive repair benefit.

In the judgment-sensitive cohort, replacing the explicit judgment changes native-30005-g0-d1 from a menu to an offer and native-30016-g5-d1 from a menu to pass, with no change in expected utility. At native-30018-g5-d2, it changes pass to an offer and increases regret from zero to $2/3$. These are observed effects of the supplied judgment on the same planner, not evidence that erroneous judgments are generally preferable. The other three root actions are unchanged.

For the forced action pairs, the higher-minus-lower reference utility contrast is 0.444 $[-0.222, 1.111]$; with the model choosing the second action, it is $-0.222 [-1.333, 0.833]$. These quantities include physical commitment changes and do not identify information-mediated utility effects. The auxiliary chooser/updater decomposition has ten complete tables, but replacing the updater changes reference-continuation utility by zero in both the utility-optimal and model-selected first-action arms in every game. We do not use these zeros as evidence that inference is irrelevant, or claim a positive updater-repair benefit. One root reference-planner score under a wrong model judgment is undefined because the assumed type contradicts the public history; that secondary score is marked unavailable, not filled with zero. Direct model action regret and the paired model-planner intervention remain defined.

Relation to the preceding native diagnosis. An earlier native run covered 36 positions from 26 source games, with known, unknown/relevant and unknown/irrelevant conditions. Its six designated confirmation unknown/relevant primary cases had B exact agreement 16.7% $[0.0, 50.0]\%$ and reference-judgment P regret 0.667 $[0.250, 1.083]$. Those primary cases mixed root and post-interaction decisions. That run produced 301 valid submissions out of 302 tasks and passed all four semantic controls. We retain it as preceding B/P evidence, without pooling it with this supplement: nine positions are reused, and the additional position is another query/history from an already represented source game. The earlier restricted three-stage protocol, which changed action structure and coupled partner types, is not the source of any results reported here.

B APPENDIX

C SUBMISSION OF CONFERENCE PAPERS TO ICLR 2027

ICLR requires electronic submissions, processed by <https://openreview.net/>. See ICLR’s website for more instructions.

If your paper is ultimately accepted, the statement `\iclrfinalcopy` should be inserted to adjust the format to the camera ready requirements.

The format for the submissions is a variant of the NeurIPS format. Please read carefully the instructions below, and follow them faithfully.

C.1 STYLE

Papers to be submitted to ICLR 2027 must be prepared according to the instructions presented here.

Authors are required to use the ICLR $\LaTeX$ style files obtainable at the ICLR website. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for rejection.

C.2 RETRIEVAL OF STYLE FILES

The style files for ICLR and other conference information are available online at:

<http://www.iclr.cc/>

The file `iclr2027_conference.pdf` contains these instructions and illustrates the various formatting requirements your ICLR paper must satisfy. Submissions must be made using $\LaTeX$ and the style files `iclr2027_conference.sty` and `iclr2027_conference.bst` (to be used with $\LaTeX$ 2e). The file `iclr2027_conference.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in sections D, E, and F below.

D GENERAL FORMATTING INSTRUCTIONS

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors' names are set in boldface, and each name is placed above its corresponding address. The lead author's name is to be listed first, and the co-authors' names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section F regarding figures, tables, acknowledgments, and references.

There will be a strict upper limit of **9 pages** for the main text of the initial submission, with unlimited additional pages for citations. This limit will be expanded to **10 pages** for rebuttal/camera ready.

E HEADINGS: FIRST LEVEL

First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

E.1 HEADINGS: SECOND LEVEL

Second level headings are in small caps, flush left and in point size 10. One line space before the second level heading and 1/2 line space after the second level heading.

E.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

F CITATIONS, FIGURES, TABLES, REFERENCES

These instructions apply to everyone, regardless of the formatter being used.

F.1 CITATIONS WITHIN THE TEXT

Citations within the text should be based on the `natbib` package and include the authors' last names and year (with the "et al." construct for more than two authors). When the authors or the publication are included in the sentence, the citation should not be in parenthesis using `\citet{}` (as in "See ? for more information."). Otherwise, the citation should be in parenthesis using `\citep{}` (as in "Deep learning shows promise to make progress towards AI (?).").

The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

F.2 FOOTNOTES

Indicate footnotes with a number¹ in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).²

¹Sample of the first footnote

²Sample of the second footnote

Table 5: Sample table title

PART	DESCRIPTION
Dendrite	Input terminal
Axon	Output terminal
Soma	Cell body (contains cell nucleus)

F.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

Make sure the figure caption does not get separated from the figure. Leave sufficient space to avoid splitting the figure and figure caption.

You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

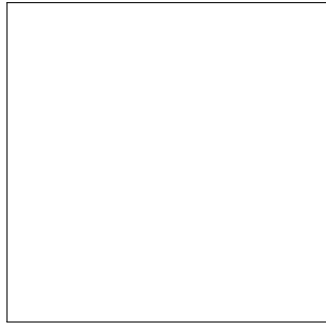


Figure 1: Sample figure caption.

F.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 5.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

G DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* ? available at https://github.com/goodfeli/dlbook_notation/. Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

Numbers and Arrays

702	a	A scalar (integer or real)
703	$\mathbf{a}$	A vector
704	$\mathbf{A}$	A matrix
705	$\mathbf{A}$	A tensor
706	$\mathbf{I}_n$	Identity matrix with n rows and n columns
707	$\mathbf{I}$	Identity matrix with dimensionality implied by context
708	$\mathbf{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
709	$\text{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by $\mathbf{a}$
710	$\mathbf{a}$	A scalar random variable
711	$\mathbf{a}$	A vector-valued random variable
712	$\mathbf{A}$	A matrix-valued random variable
713	Sets and Graphs	
714	$\mathbb{A}$	A set
715	$\mathbb{R}$	The set of real numbers
716	$\{0, 1\}$	The set containing 0 and 1
717	$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
718	$[a, b]$	The real interval including a and b
719	$(a, b]$	The real interval excluding a but including b
720	$\mathbb{A} \setminus \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
721	$\mathcal{G}$	A graph
722	$\text{Pa}_{\mathcal{G}}(\mathbf{x}_i)$	The parents of $\mathbf{x}_i$ in $\mathcal{G}$
723	Indexing	
724	a_i	Element i of vector $\mathbf{a}$, with indexing starting at 1
725	$\mathbf{a}_{-i}$	All elements of vector $\mathbf{a}$ except for element i
726	$\mathbf{A}_{i,j}$	Element i, j of matrix $\mathbf{A}$
727	$\mathbf{A}_{i,:}$	Row i of matrix $\mathbf{A}$
728	$\mathbf{A}_{:,i}$	Column i of matrix $\mathbf{A}$
729	$\mathbf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathbf{A}$
730	$\mathbf{A}_{:,:,i}$	2-D slice of a 3-D tensor
731	$\mathbf{a}_i$	Element i of the random vector $\mathbf{a}$

Calculus

756	$\frac{dy}{dx}$	Derivative of y with respect to x
757	$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
758	$\nabla_{\mathbf{x}} y$	Gradient of y with respect to $\mathbf{x}$
759	$\nabla_{\mathbf{X}} y$	Matrix derivatives of y with respect to $\mathbf{X}$
760	$\nabla_{\mathbf{x}}^2 y$	Tensor containing derivatives of y with respect to $\mathbf{x}$
761	$\frac{\partial f}{\partial \mathbf{x}}$	Jacobian matrix $\mathbf{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
762	$\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$	The Hessian matrix of f at input point $\mathbf{x}$
763	$\int f(\mathbf{x}) d\mathbf{x}$	Definite integral over the entire domain of $\mathbf{x}$
764	$\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$	Definite integral with respect to $\mathbf{x}$ over the set $\mathbb{S}$
765		
766		Probability and Information Theory
767	$P(\mathbf{a})$	A probability distribution over a discrete variable
768	$p(\mathbf{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
769	$\mathbf{a} \sim P$	Random variable $\mathbf{a}$ has distribution P
770	$\mathbb{E}_{\mathbf{x} \sim P}[f(\mathbf{x})]$ or $\mathbb{E}f(\mathbf{x})$	Expectation of $f(\mathbf{x})$ with respect to $P(\mathbf{x})$
771	$\text{Var}(f(\mathbf{x}))$	Variance of $f(\mathbf{x})$ under $P(\mathbf{x})$
772	$\text{Cov}(f(\mathbf{x}), g(\mathbf{x}))$	Covariance of $f(\mathbf{x})$ and $g(\mathbf{x})$ under $P(\mathbf{x})$
773	$H(\mathbf{x})$	Shannon entropy of the random variable $\mathbf{x}$
774	$D_{\text{KL}}(P \ Q)$	Kullback-Leibler divergence of P and Q
775	$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$
776		
777		Functions
778	$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
779	$f \circ g$	Composition of the functions f and g
780	$f(\mathbf{x}; \boldsymbol{\theta})$	A function of $\mathbf{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\mathbf{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
781	$\log x$	Natural logarithm of x
782	$\sigma(x)$	Logistic sigmoid, $\frac{1}{1 + \exp(-x)}$
783	$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
784	$\ \mathbf{x}\ _p$	L^p norm of $\mathbf{x}$
785	$\ \mathbf{x}\ $	L^2 norm of $\mathbf{x}$
786	x^+	Positive part of x , i.e., $\max(0, x)$
787	$\mathbf{1}_{\text{condition}}$	is 1 if the condition is true, 0 otherwise
788		
789		
790		
791		
792		
793		
794		
795		
796		
797		
798		
799		
800		
801		
802		
803		
804		
805		
806		
807		
808		
809		

H FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

I PREPARING POSTSCRIPT OR PDF FILES

Please prepare PostScript or PDF files with paper size “US Letter”, and not, for example, “A4”. The `-t letter` option on `dvips` will produce US Letter files.

Consider directly generating PDF files using `pdflatex` (especially if you are a MiKTeX user). PDF figures must be substituted for EPS figures, however.

Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

I.1 MARGINS IN LATEX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below using `.eps` graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for `.pdf` graphics. See section 4.4 in the graphics bundle documentation (<http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps>)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command.